

Welcome to KSR Datavizon



Data ware housing

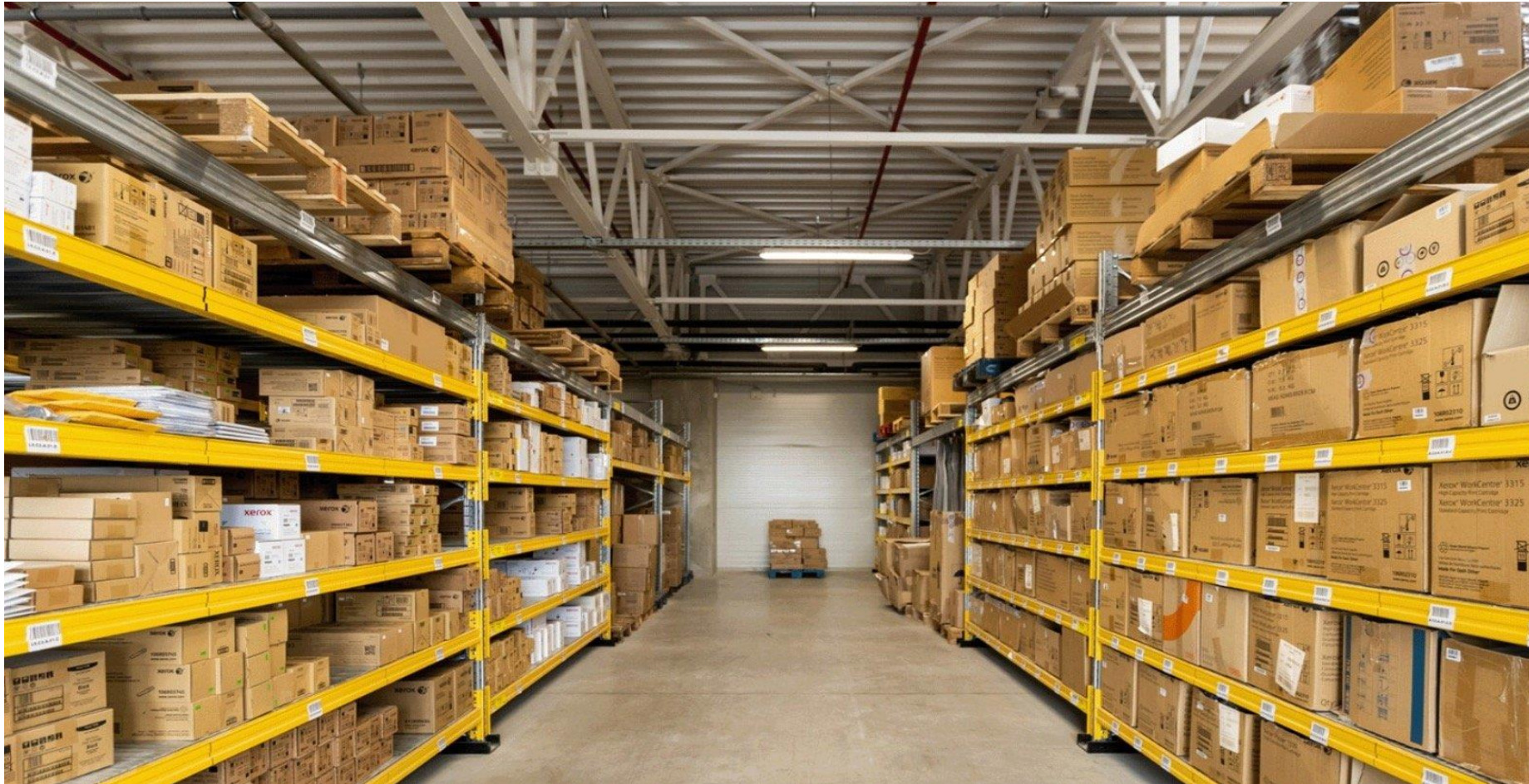


Agenda

- What is data ware housing?
- Why it is needed?
- ETL
- ETL vs ELT
- Dataware vs DataMart
- Dimension / Fact tables
- SCDs
- On prem ETL Tools



Yeah it's a warehouse !!!

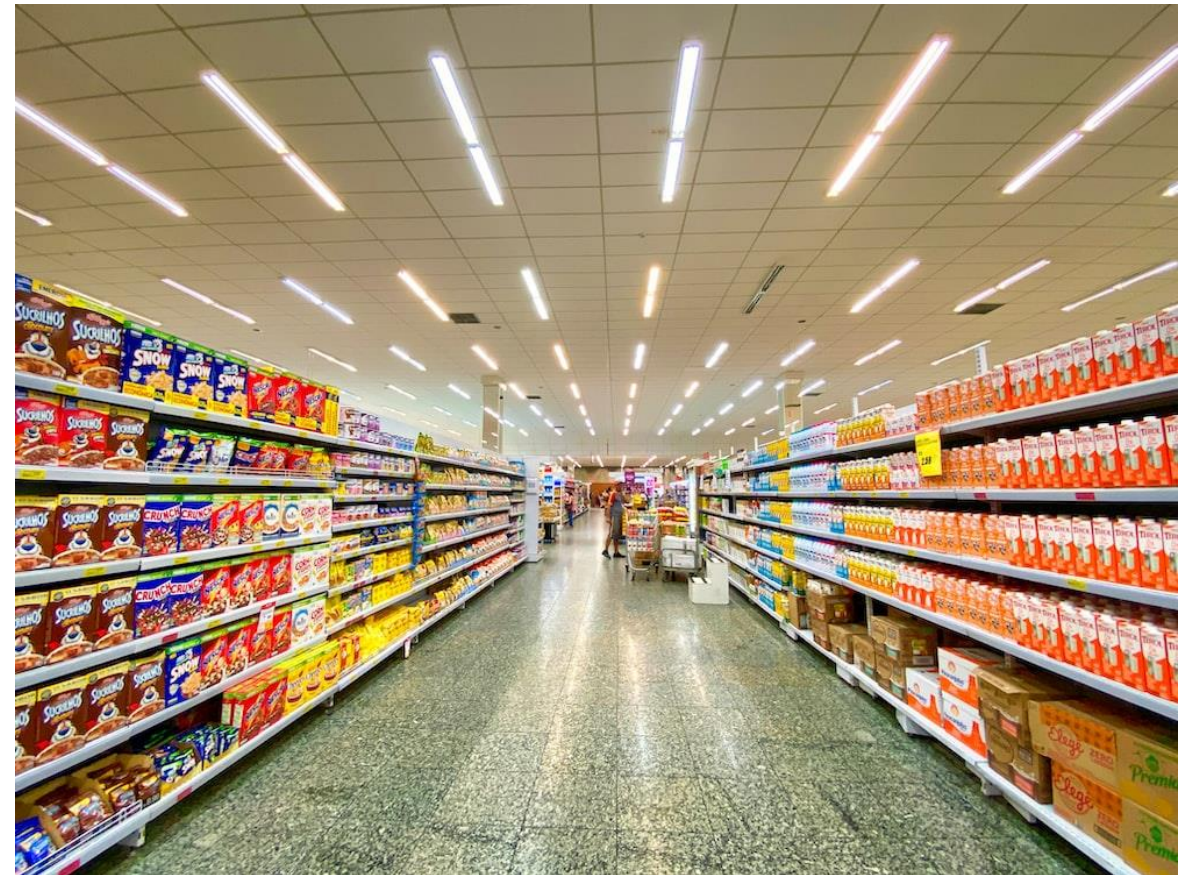


A warehouse is a large commercial storage building used to store goods, products, and materials. Warehouses are typically used by businesses and organizations to store inventory, raw materials, and finished products for distribution or sale



Why it is needed?

- Data is spread across different systems (spreadsheet, sql server, csv)
- Useful for reporting
- One stop shop (for all details)
- Historical data
- Performance
- Decision making
- Data Consistency

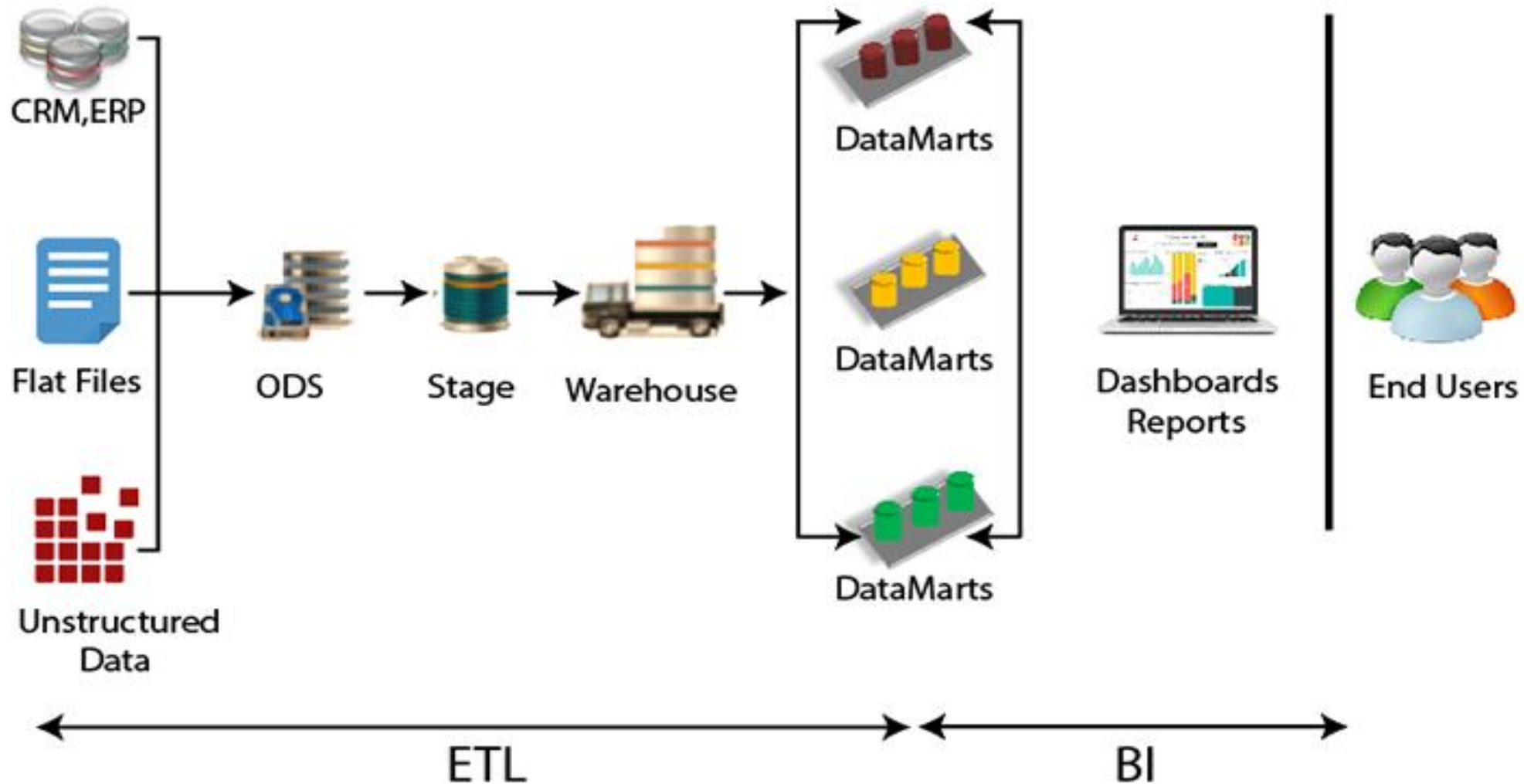


Database vs data warehouse

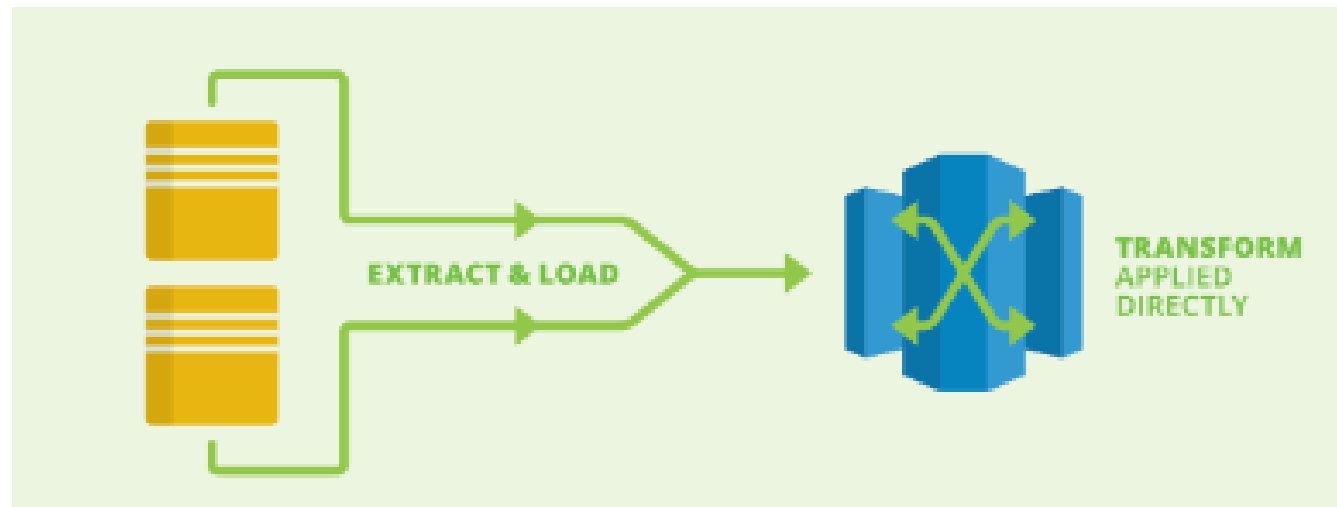
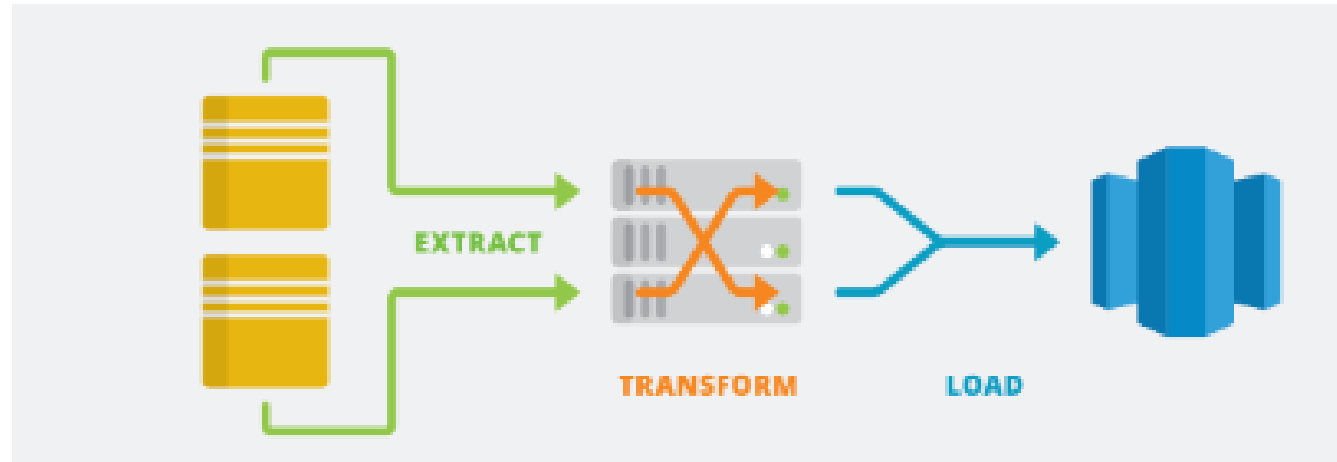
	Database	Data warehouse
Processing type	OLTP	OLAP
Optimized operations	CRUD transactions	Complex analytical queries
Data sources	Usually one	Usually multiple
Data model	Normalized	Denormalized
Data timelines	Daily to monthly	Historical
Data volume	Low to mid	Mid to large



ETL Process



ETL vs ELT



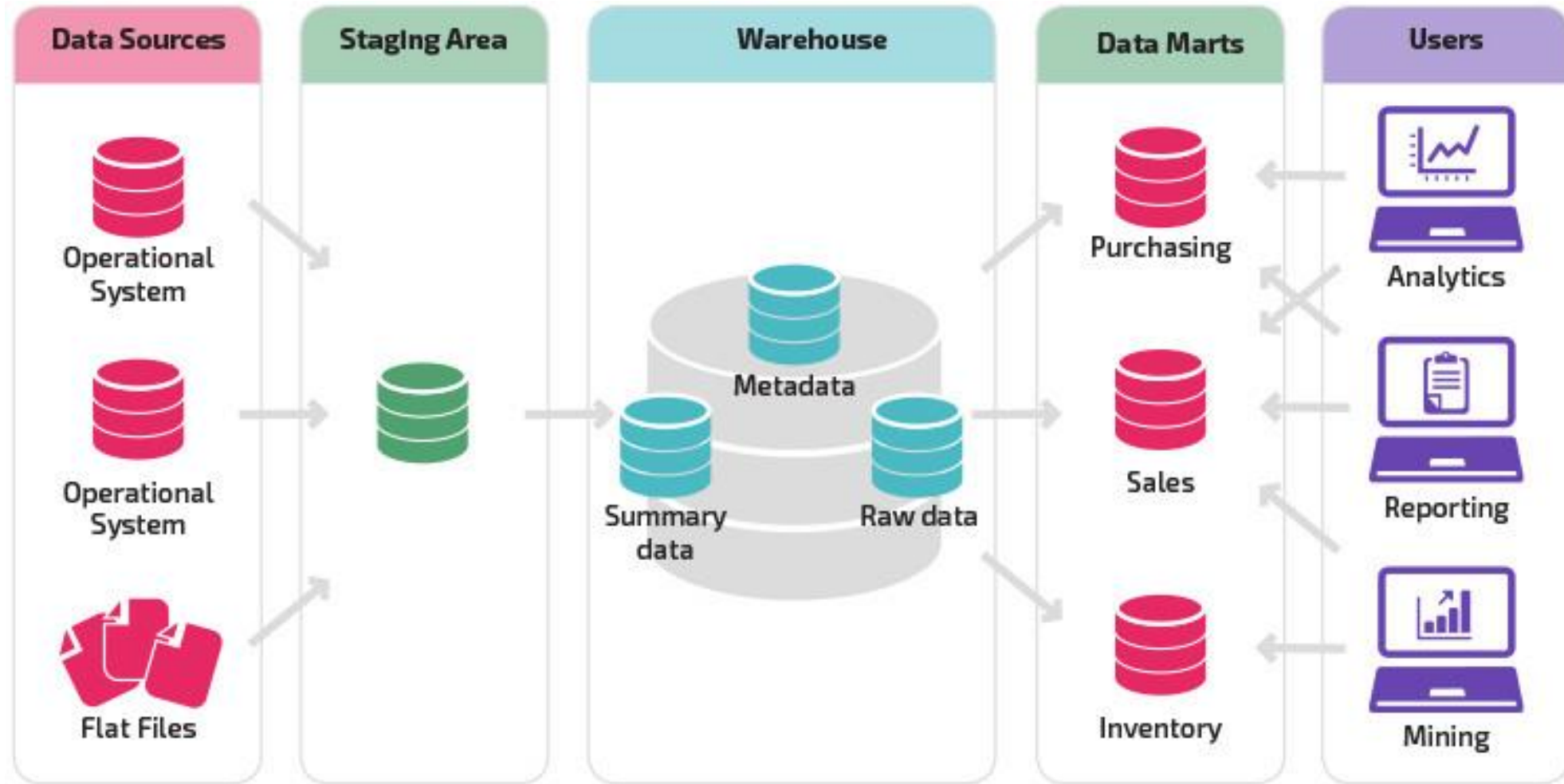
ETL



ELT



Dataware house vs DataMart



2 Types of ETL

- Initial ETL
 - One time
 - Right before dataware goes live
 - Bringing all relevant information
- Incremental ETL
 - Incrementally refreshes
 - New data – new customer, products
 - Modified data – update customer details



ETL Tools

IBM Infosphere Information Server

Fast track your information



 **talend**

6.4



**Talend Open Studio
for Data Integration**

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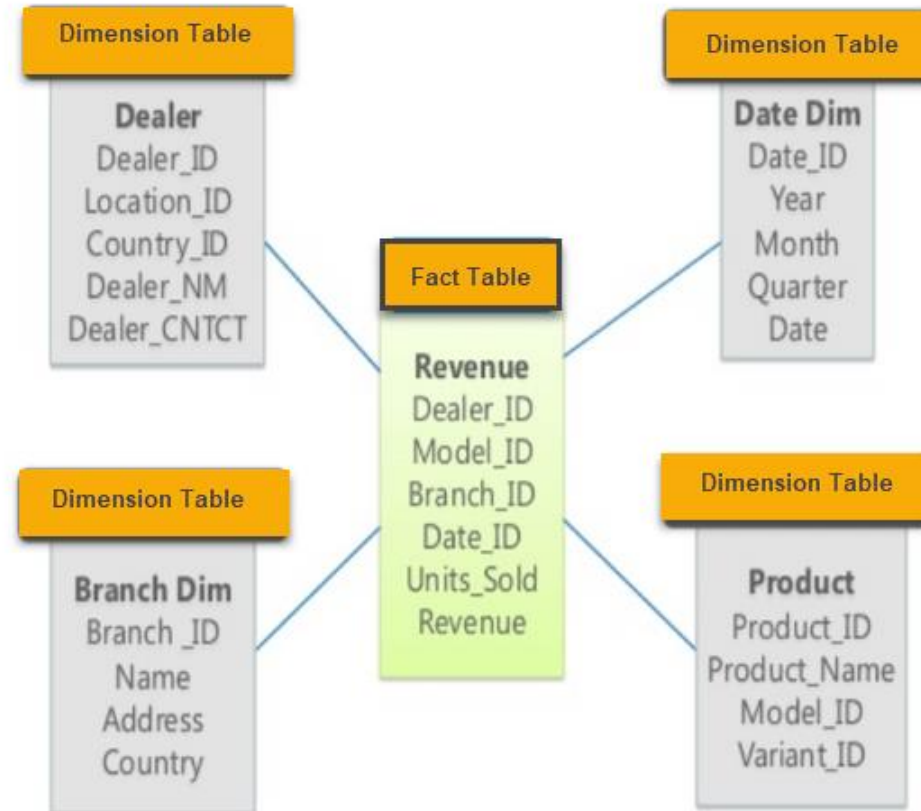


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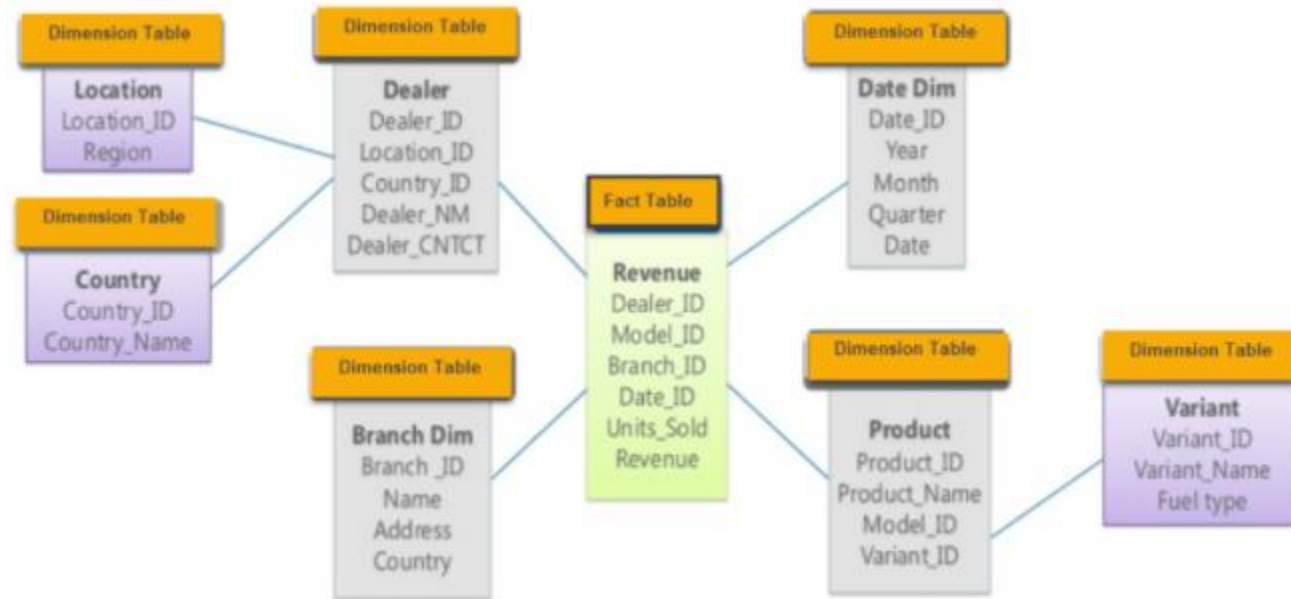
Star Schema



Example of Star Schema Diagram



Snowflake Schema



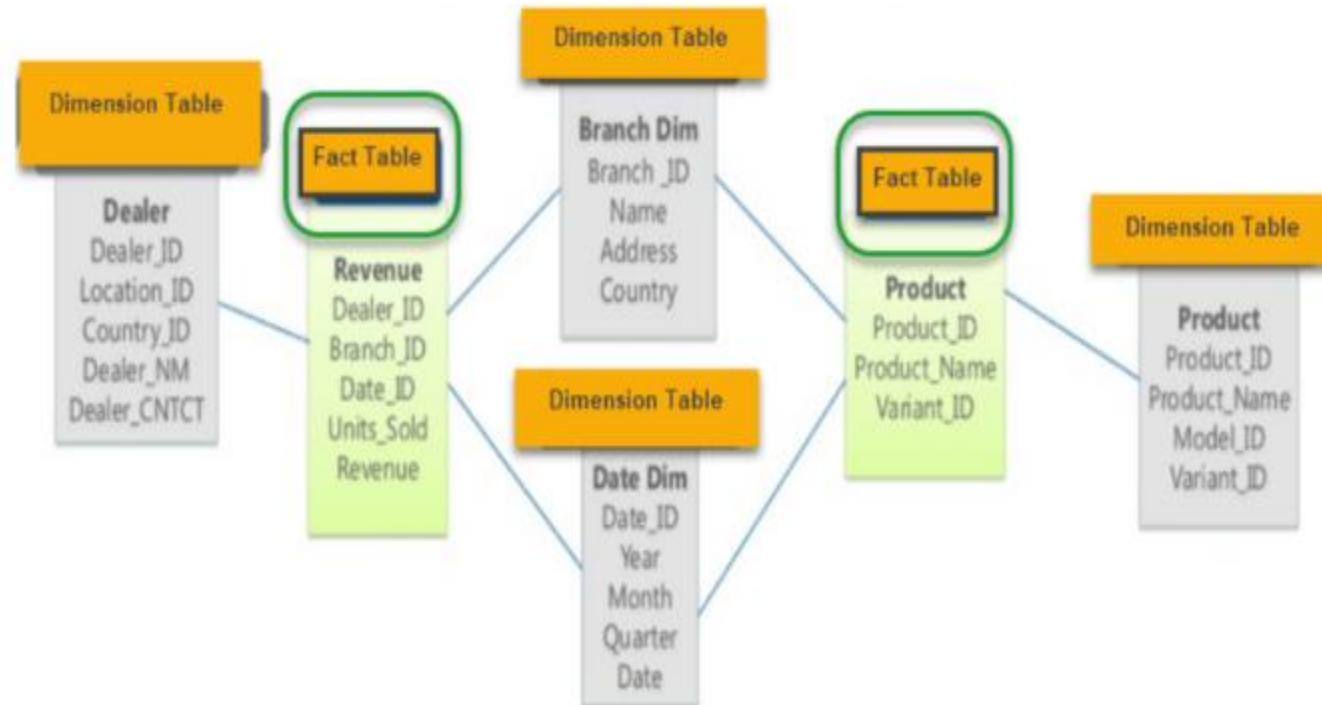
Example of Snowflake Schema



Star Schema	Snowflake Schema
Hierarchies for the dimensions are stored in the dimensional table.	Hierarchies are divided into separate tables.
It contains a fact table surrounded by dimension tables.	One fact table surrounded by dimension table which are in turn surrounded by dimension table
In a star schema, only single join creates the relationship between the fact table and any dimension tables.	A snowflake schema requires many joins to fetch the data.
Simple DB Design.	Very Complex DB Design.
Single Dimension table contains aggregated data.	Data Split into different Dimension Tables.
Offers higher performing queries using Star Join Query Optimization. Tables may be connected with multiple dimensions.	The Snowflake schema is represented by centralized fact table which unlikely connected with multiple dimensions



Galaxy / Bridge Schema



Example of Galaxy Schema



Questions ???



Thank You!!!





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